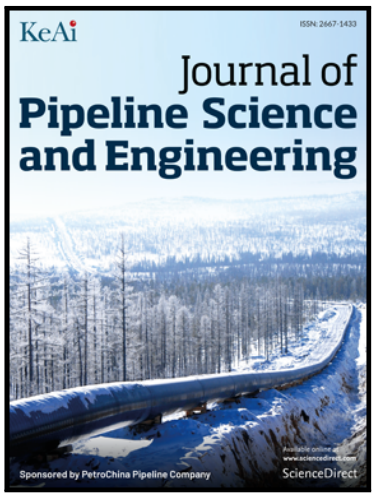
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Prediction of Internal Corrosion in Oil and Gas Piping Systems—An Independent Analysis Using Machine Learning

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Abstract

This work presents machine learning (ML) algorithm development and testing for internal corrosion rate prediction in oil and gas (O&G) piping systems. The ML models were trained using TWI's dataset of environmental and fluid parameters, operational and flow parameters, as well as pipeline and material parameters at 35000+ condition monitoring locations (CMLs). We have trained two ML models based on different artificial intelligence (AI) approaches: model 1, a direct regression model, and model 2, a composite classification-regression model. The ML models were created using data that posed remarkable challenges, such as biasedness, skewness, and duplicate values. We observed better statistical metrics for predicting using model 2 compared to prediction using model 1. Systematic parameter selection, power transformation, and hyperparameter optimization using the grid search technique were employed in the model development. Although the model 2 predicted well for lower corrosion rate (0.05mm/year), it fell short in predicting accurately for higher corrosion rates (>0.2mm/year) as the uncertainty increases at higher corrosion rates. We have also provided a discussion on the uncertainty in the model, targeted performance metrics, use of correction factors, improving model accuracy through model refinement, and reducing bias nature of the dataset by adding more diverse values at higher corrosion rates.

Keywords: Internal corrosion, oil and gas, piping systems, machine learning.

1. Introduction

1.1. Study Rationale

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"Oil and gas piping" refers to a system of interconnected pipes, fittings, and valves designed to transport crude oil, natural gas, and related petroleum products from production sites to refineries and distribution networks, adhering to specific American Petroleum Institute (API) standards regarding materials, design, and construction to ensure safe and reliable operation within the O&G industry; whereas "oil and gas pipeline" refers to a system of pipes used to transport crude oil and natural gas from extraction sites to refineries or consumer markets over long distances. As pipelines are built for long distance transportation, they have large diameter pipes with fewer fittings. On the other hand, piping systems have numerous variations in the fittings to handle the O&G flow in the complex refinery setup. The fluids transported by pipelines are mainly non-corrosive liquids and dry gas while piping handles a variety of corrosive fluids and in different caustic environments. As the accessibility of piping system is always challenging, and with offshore platforms translating towards unmanned and remote operations, the piping inspection is tedious compared to the pipelines. Standards such as American Society of Mechanical Engineers (ASME) B31.3 [1] elaborate the design code for piping system whereas API 570 [2] specifically outlines the piping inspection code. Thus, the research work on O&G piping systems must be distinguished from the O&G pipeline studies as there are differences in dimensions, flow parameters, operating conditions, etc. between them.

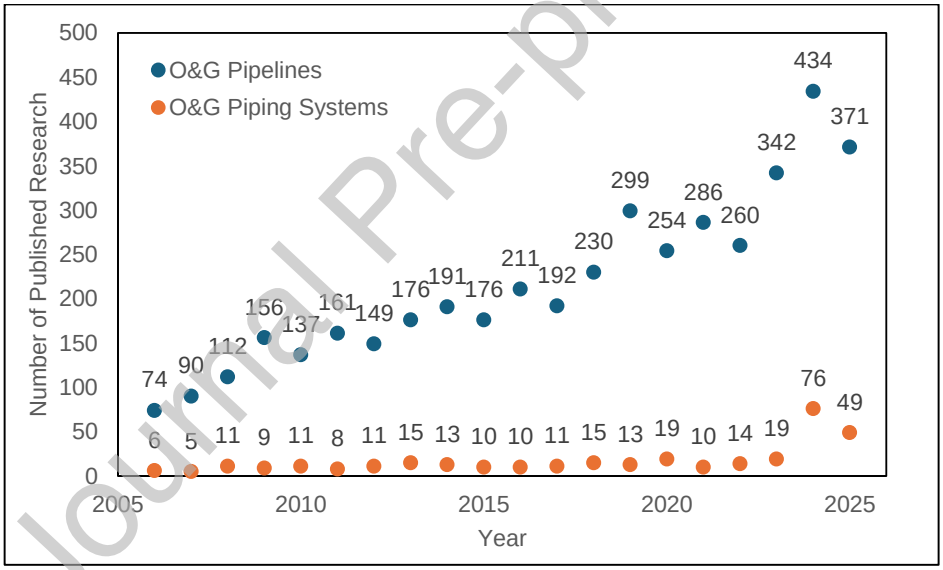


Figure 1. Scopus search results for “Corrosion in oil and gas pipelines” and “Corrosion in oil and gas piping systems”

In Scopus database, an advanced search of “Corrosion in oil and gas pipelines” produces 4301 literatures during 2005-2025 whereas “Corrosion in oil and gas piping” produces only 335 literatures during the same period (Figure 1). To the authors knowledge, there is no existing literature on the use of ML for prediction of internal corrosion in piping systems. All the existing studies on AI assisted corrosion monitoring in the O&G sector are conducted using pipeline variables thereby leaving a gap for work to be done on piping systems. Despite their high accuracy, advanced ML models for corrosion prediction in piping systems find less application in the academia (due to lack of data) and in the industry (due to more dependability on empirical models). By merging the insights from academia and industry, the

study aims to develop an independent ML model for predicting corrosion rates of piping systems using TWI's dataset of corrosion measurements.

1.2. Background

Corrosion is generally defined as the deterioration of the metal surface due to its chemical interaction with the surrounding. In petro-chemical industries, the surrounding atmosphere combined with the highly reactive organic compounds, the methodology of extracting petroleum and transportation of its products together create an environment that propagates corrosion. Numerous chemical and physical factors are responsible for corrosion in the O&G industry. H_2S, CO_2 and chloride ions are major chemical contaminants while operating conditions such as temperature, pressure, humidity, acidity, salinity, flow rate, deposits and scales are the factors held responsible for internal corrosion in all the industrial components, giving rise to high probability of failure. The dynamic nature of these factors makes the finance and safety of the industries complex.

Internal corrosion in metal pipes forms the cause of operational damage within a plant such as breakdown, turmoil in production and environmental catastrophe. The highly resistant nature of metal pipes to high temperature and pressure makes it suitable for fast, safe and economic use in refining industries [3]. They hold a huge significance in transporting petroleum products that form one of the world's most important energy resources. However, metal pipes are vulnerable to leaks caused mainly due to corrosion and reduce their lifespan [4]. The degradation and corrosion of metal pipes is mainly due to the impurities carried along the O&G pipes—carbon dioxide (CO_2), hydrogen sulfide (H_2S), water (H_2O) and microbes. The consequence of these impurities along with the operating temperature and pressure is internal corrosion in the O&G piping systems.

Crude O&G contain impurities that are caustic in nature. During transportation of these fluids using the pipe, the inner surface of the metal pipe deteriorates which is known as internal corrosion [5,6]. Internal corrosion in pipes leads to flaws such as pits and ruptures, thereby depleting the material of the pipe which eventually escalates the threat of system collapse [7]. To have a safe and reliable operation of the piping systems in the petroleum industry, a good knowledge and calculations of failure mechanisms involved, and their respective risk factors is mandatory. Prediction of the internal corrosion rates and then applying the precautionary techniques accordingly reduces the probability of failure of the components and it forms a core segment of asset integrity assessment and management. Priyanka and Thangavel [8] list that around 33% of metal pipe mishaps are consequences of inappropriate prediction of the corrosion rate emphasizing the need of accurate techniques that can reduce economic losses and save human life from accidents.

To reduce the severity of the consequences of corrosion, a master plan is a must in which corrosion prediction holds utmost importance. Any ignorance in mitigating the corrosion may result in breakdown, spill or rupture of piping system and other components eventually leading to high-price restoration, reduction in productivity, pollution and in the worst circumstances, loss of lives. Thus, it becomes mandatory to maintain the asset integrity for the welfare and benefits of the industry by timely inspection of the corrosion and reducing its damage.

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In many different industries, including the O&G, corrosion is one of the main problems affecting the integrity and safety of piping systems. Manual inspection and review are used in traditional methods of assessing corrosion; these processes are time-consuming, subjective, and frequently inaccurate. By creating predictive models based on data from various sources, it is now feasible to increase the accuracy and efficiency of corrosion assessment thanks to the development of ML and AI. To forecast the rate of corrosion and remaining life of the piping system based on the input data, ML models can be trained using historical datasets. This makes it possible to plan repair and replacement tasks more proactively, lowering the likelihood of failure and downtime. Industry adoption of ML for piping system corrosion assessment is still in its infancy, and there are still difficulties to be overcome, including the need for vast quantities of high-quality data, the complexity of the models, and the requirement for expert knowledge to interpret the findings. ML, however, has the potential to revolutionize how industry assesses and manages corrosion in piping systems and other assets as more data becomes accessible and the models become more complex.

The advancement in AI has contributed significantly to the field of asset integrity management. The application of AI for metal pipe corrosion is spread widely across the fields of science [9]. The existing literatures validate that people working in the domain of condition monitoring are extensively using ML algorithms as these algorithms tend to predict accurately and are proving themselves effective [9,10]. But it is remarkable to note that one algorithm doesn't fit all the conditions as different pipelines and piping systems may be exposed to different operating conditions [11]. Also, each algorithm has its own advantages and limitations, so it is important to choose the better AI technique depending on the operating conditions and environment for precise corrosion detection. Support vector regression (SVR) algorithm, artificial neural network (ANN), ensemble learning (EL) are some of the advanced AI techniques that are successful in predicting the corrosion rate and depth with a high accuracy [12,13]. Altogether, it can be deduced that AI is proving to be a functional aid across asset integrity management and monitoring systems, especially in the Petro-chemical sector where metal pipes which are prone to corrosion are spread in millions of kilometers.

The API publishes a wide number of recommended practice standards that provides the most widely used guidelines in the refining industry for design, manufacture and testing of components. The present work considers the recommendations from API 571 [14] for concentrating on the damage mechanisms as well as recommendations from API 580 [15] and API 581 [16] for considering the risk-based inspection and risk-based inspection methodology respectively. This consideration helps to minimize the gap between industrial practices and academic research. Understanding of the factors responsible for internal corrosion is mandatory for cleaning of the data and having an effective machine learning prediction.

1.3. Objective of the Study

The objective of this study is to develop an independent ML model for predicting corrosion rates of piping systems using TWI's dataset of corrosion measurements. Two ML models were developed using regression and classification routes, Model 1: A direct regression model and Model 2: A composite classification-regression model. The dataset contained 35000+ CMLs, each had both numerical and

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categorical features. For both Model 1 and Model 2, four regression algorithms were tested, whereas for Model 2, four classification training algorithms were investigated, to find out the algorithm having the best performance for the given dataset. Further optimization of hyperparameters using the grid search method was also conducted to achieve the highest possible accuracy for selected features.

2. Material and Methods

The following subsections discuss the details of the dataset and the methodology used for model development. All data processing and statistical analyses were performed using Python (version 3.9) with the following libraries: pandas (version 1.5.3) for data manipulation, numpy (version 1.24.3) for numerical computations, matplotlib (version 3.7.1) and seaborn (version 0.12.2) for data visualization, and scikit-learn (version 1.2.2) for statistical transformations.

2.1. Understanding the dataset

Historical data on refinery piping system available with TWI's Software and Engineering Services (SES) team has 35160 datasets with 52 features. The parameters of these 35160 entries included fluid type, unit name, operating temperature (℃), operating pressure (bar), design temperature (℃), design pressure (bar), commissioning date of piping system, material type, material grade, CML no., nominal pipe size (inch), nominal thickness (mm), minimum measured thickness (mm), minimum required thickness (mm), local corrosion rate (mm/year), probabilistic corrosion rate, targeted corrosion rate, remaining life (year), last inspection date, date of inspection before last inspection, component type, containing dead leg (yes/no), fluid phase, outside diameter (mm), inside diameter (mm), liquid flow rate (barrels/day), gas flow rate (m^3/day), liquid fluid velocity (m/s), gas fluid velocity (m/s), fluid velocity (m/s), viscosity (CP), density (kg/m^3), Reynolds number, the site is upstream or downstream process, is there sand in the fluid (yes/no), internal corrosion-erosion (yes/no), chemical injection (yes/no), highly effective inspection for corrosion at injection point according to API 570 (yes/no), mixing point (yes/no), cyclic loading connected within 15.25m (yes/no), water content (%), dew point temperature (℃), saturation pressure (bar), relative humidity (%), free water (yes/no), water vapor pressure (bar), mole fraction CO_2, mole fraction H_2S, partial pressure CO_2 (bar), partial pressure H_2S (bar), pH (CO_2 + H_2S) and aging (days).This dataset has been made available to TWI through two partner offshore facilities.

To prevent overfitting due to low bias and high variance [17] as well as to prevent underfitting due to high bias and low complexity [18], the dataset was narrowed down to 13 features for building the two models (12 independent variables and a target variable). The selection of independent variables was carried out as per the suggestions in the existing studies of corrosion in oil industry and corrosion engineering. The internal corrosion rate is highly dependent on water chemistry (pH and Cl^- content), CO_2 partial pressure, H_2S partial pressure, operating temperature and pressure, flow velocity [19,20]. Advanced models of corrosion prediction indicate that corrosion is dependent on a nexus of environmental, fluid chemical, operational, physical, material and structural parameters [11]. In the present work, the 13 selected variables consist of environmental and fluid chemical parameters, operational and physical parameters as well as material parameters and structural parameters. For predicting local corrosion rate (target variable), the independent variables selected in the dataset are

operating temperature, operating pressure, minimum measured thickness, aging of the piping system in days (last inspection date – commissioning date of piping system), inside diameter, water content, pH (CO_2 + H_2S), fluid type, material type, material grade, component type and fluid phase. Out of the 13 variables, 8 of them, namely operating temperature, operating pressure, minimum measured thickness, aging (days), inside diameter, water content, pH (CO_2 + H_2S) and local corrosion rate are numerical data. As the present study is an independent analysis, the authors do not claim that the selected independent variables are the only factors that must be accounted for predicting corrosion rate. Table 1 shows mean, median, mode, standard deviation (SD), minimum value (Min.) and maximum value (Max.) of these variables. The statistical characteristics of the numerical dataset indicated high degree of variability in the data, such as in operating temperature, operating pressure, minimum measured thickness, aging, inside diameter, water content, and pH. The distribution of local corrosion rate values showed that a significant amount of data had a corrosion rate of 0.05 mm/year, which could affect the model's accuracy and uncertainty in the model.

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Table 1. Descriptive statistics of numerical variables in the dataset

Index	Operating Temperature	Operating Pressure	Minimum Measured Thickness	Aging	Inside Diameter	Water Content	pH (CO_2+H_2S)	Local Corrosion Rate
Count	35160	35160	35160	35160	35160	35160	35160	35160
Mean	41.33	26.44	10.09	9488.5	201.1	23.34	4.677852	0.070108
Median	37	13.7	8.2	9101	146.36	1.57	4.190034	0.05
Mode	25	1	8	12265	42.82	0.01	7.113412	0.05
SD	30.82	66.78	7.15	4595.5	189.3	31.95	1.264732	0.054616
Min.	-17.8	0.08	1.1	0	-0.84	0	2.966061601	0.05
Max.	405	390	88.3	20789	1497	99.99	7.94543939	1.7

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The remaining 5 variables in the dataset; fluid type, material type, material grade, component type and fluid phase are categorical features. Table 2 shows the total count of each of these categorical variables in the dataset.

Table 2. Tally of categorical variables. Count is the number of different types of categorical features present in the dataset

Features	Count
Fluid Type	62
Material Type	5
Material Grade	16
Component Type	8
Fluid Phase	3

2.2. Preprocessing

The dataset contained many duplicate data with the same or different targets, which could lead to confusion and errors in the model. The dataset of 13 features consisted of 35160 entries having unique and duplicate entries. The duplicate rows of data was dropped from the dataset to keep only the first occurrence of each unique feature combination. This reduced the total number of data rows of each variable to 16095, thereby removing 19065 data rows. The categorical features were then converted to numerical values using target encoding technique. Target encoding converts categorical features to numerical values using mean of the target variable for data in that variable [21].

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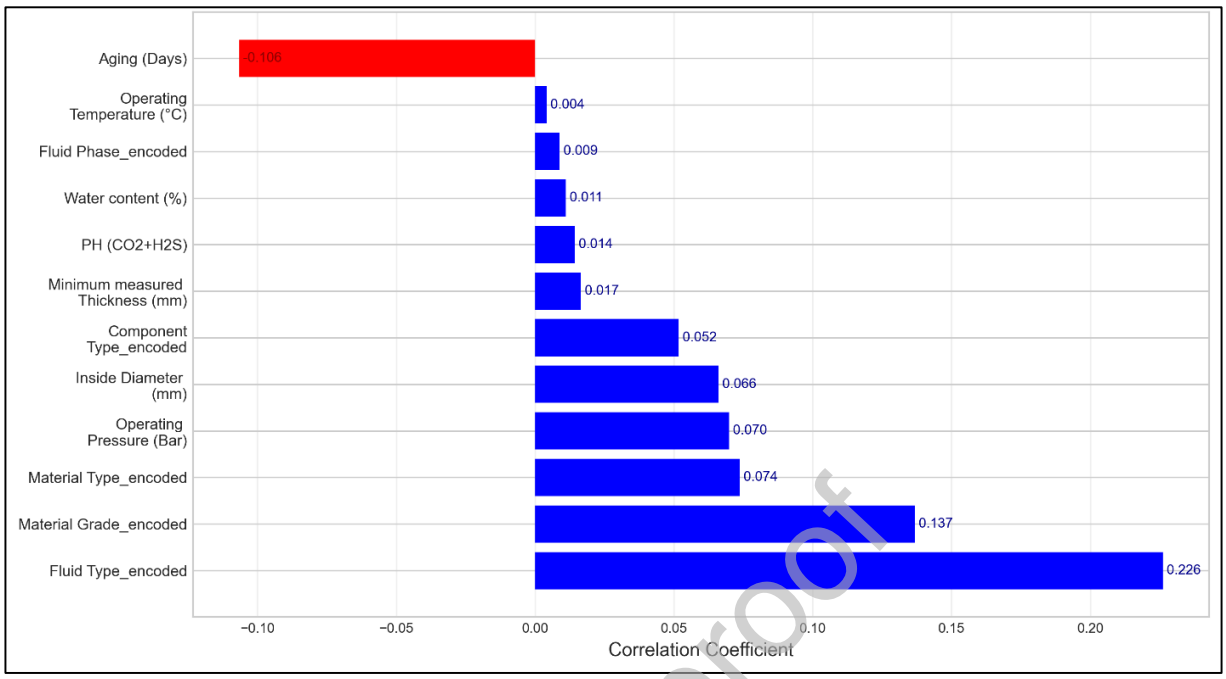


Figure 2. Correlation of independent variables with the target variable (local corrosion rate)

To understand the correlation between the target variable and independent variables, Pearson correlation analysis was conducted. Figure 2 shows Pearson correlation coefficient plot between the independent variables and the target variable. The low correlation coefficient indicates that the relation between the high relevant attributes and local corrosion rate is non-linear. The coefficient correlation shows that other than aging, fluid type and material grade, other features had very low linear correlation with the target. As multivariate models depend on weak predictors [22,23], all the variables were retained for model development.

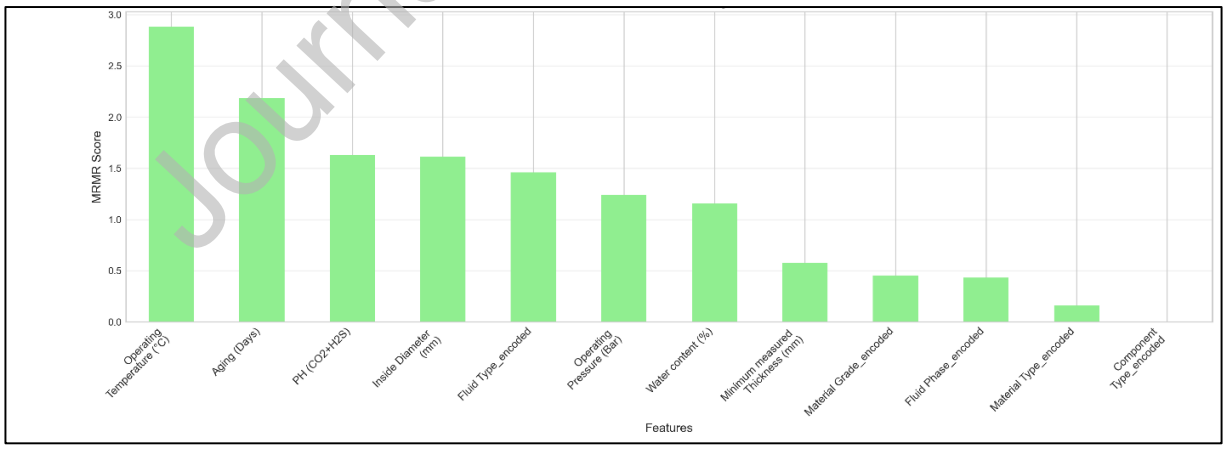


Figure 3. Feature ranking based on minimum redundancy maximum relevance (mRMR) method

The feature analysis was further carried out using minimum redundancy maximum relevance (mRMR) technique which signifies the relevance of the independent variables with the target variable while penalizing the redundancy among the independent variables [24]. Despite having a low linear correlation

with the target variable, operating temperature has the highest mRMR score (Figure 3). This shows that operating temperature provides the most unique and minimal redundant predictive data followed by aging, pH, inside diameter, fluid type, operating pressure, water content, minimum measured thickness and material grade. Material type and component type had a low mRMR score indicating that they had less unique predictive information.

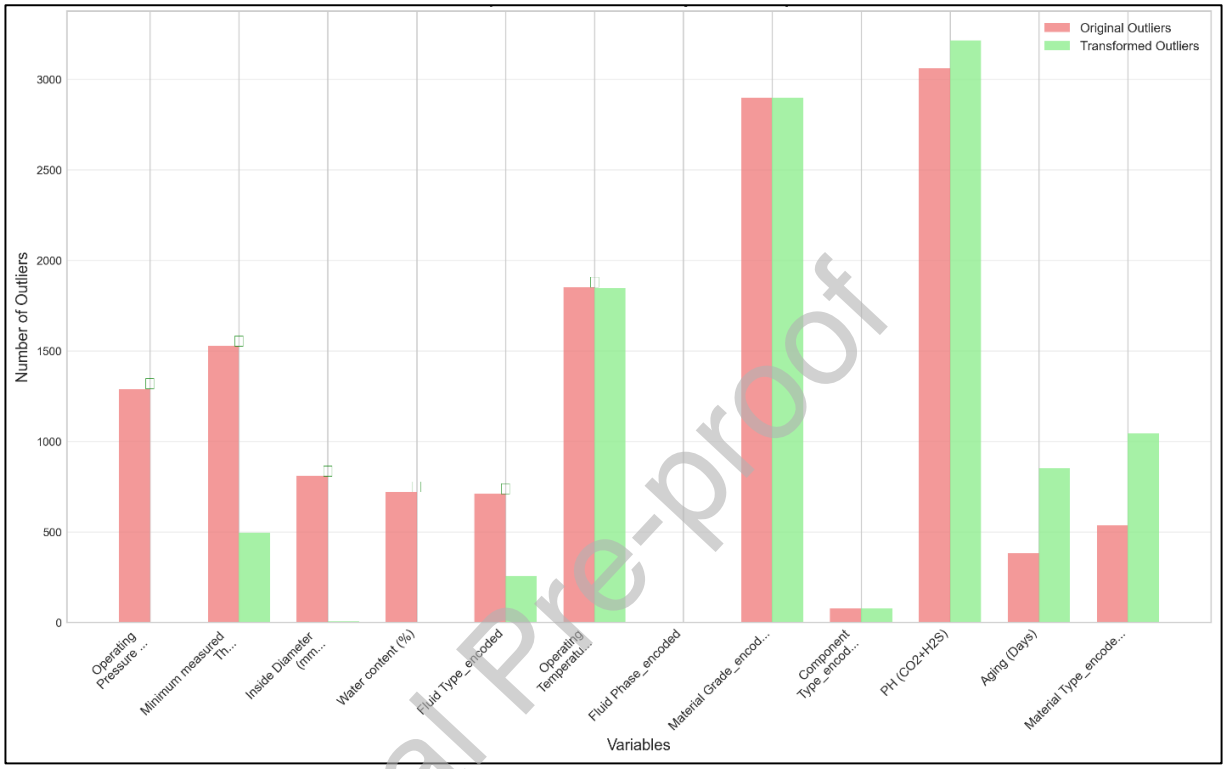


Figure 4. Outlier comparison of independent variables. Green bars show improvement

The power transformation method (Yeo-Johnson transformation) was used to reduce the number of outliers in the data. Power transformation improves the normal distribution of the data and reduces skewness. Yeo-Johnson transformation normalizes most of the data around the mean by reducing skewness, it magnifies extreme values which increases outliers in some cases. The number of outliers in the variables aging and pH increased, while it decreased significantly in operating pressure, minimum measured thickness, inside diameter and water content whereas there was few change in operating temperature and minimum measured thickness (Figure 4). Table 3 summarizes in detail the number of outliers before and after the Yeo-Johnson power transformation. After power transformation, the number of outliers decreased in the operating pressure, minimum measured thickness, inside diameter and water content. For the operating pressure and water content, the outliers were reduced to zero, while for the inside diameter, the number of outliers decreased from 808 to 7. However, the number of outliers is still high in the operating temperature and minimum measured thickness due to their asymmetric distribution. Keeping the outliers in the model may cause the model to be less accurate. Equally outliers can be removed from ML models, but this may also lead to the removal of important data, especially those related to corrosion rates higher than 0.05 mm/year. Since these outliers are legitimate extreme values, they were retained in the dataset for model development.

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Table 3. Comparison of outliers in the dataset before and after power transformation

Variable	Original Outliers	Transformed Outliers	Outlier Reduction	Reduction Percentage
Operating Temperature (°C)	1851	1846	5	0.3%
Operating Pressure (Bar)	1290	0	1290	100.0%
Minimum measured Thickness (mm)	1526	494	1032	67.6%
Aging (Days)	383	851	-468	-122.2%
Inside Diameter (mm)	808	7	801	99.1%
Water content (%)	721	0	721	100.0%
pH (CO_2+H_2S)	3060	3214	-154	-5.0%
Fluid Type	711	259	452	63.6%
Material Type	537	1043	-506	-94.2%
Material Grade	2897	2897	0	0.0%
Component Type	79	79	0	0.0%
Fluid Phase	0	0	0	N/A

2.3. Model Development

The main goal of the developed model was to predict the local corrosion rate, and the objective was to find any trends, correlations, or discrepancies in the data that would influence how well the machine learning model performed. Both regression and classification routes were followed to develop two models (Model 1 and Model 2). While both models are good enough to be used, but limitation and accuracy varies.

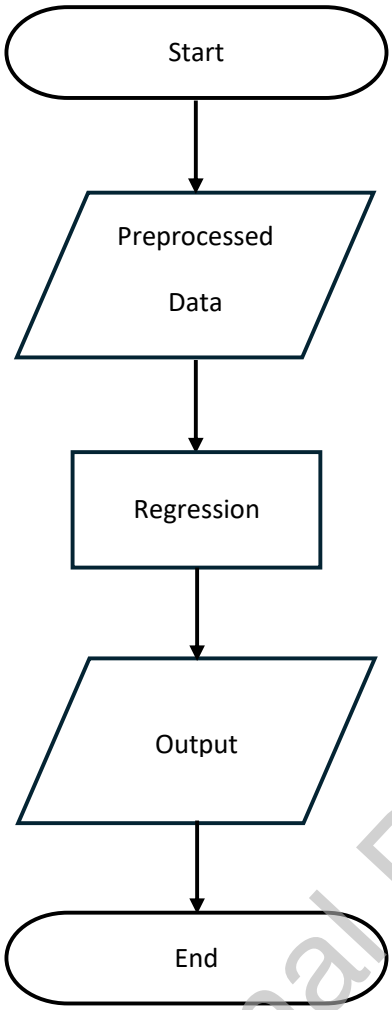
In model 1, the preprocessed data is directly fed into the regression model to predict the local corrosion rate (Figure 5a). In the regression step, the models were trained and tested using four regression techniques. The model allows us to select a regression technique out of Support vector machine (SVM), Stochastic gradient descent (SGD), Multi-layer perceptron (MLP), and Voting Regressor (SVM+MLP+SGD) algorithms.

Figure 5b is the schematic flowchart of the composite classification-regression approach for corrosion rate prediction. In model 2, initially, a classification algorithm is used to divide the data into two groups based on the target value, one with a value of 0.05 mm/year or less and the other with a value greater

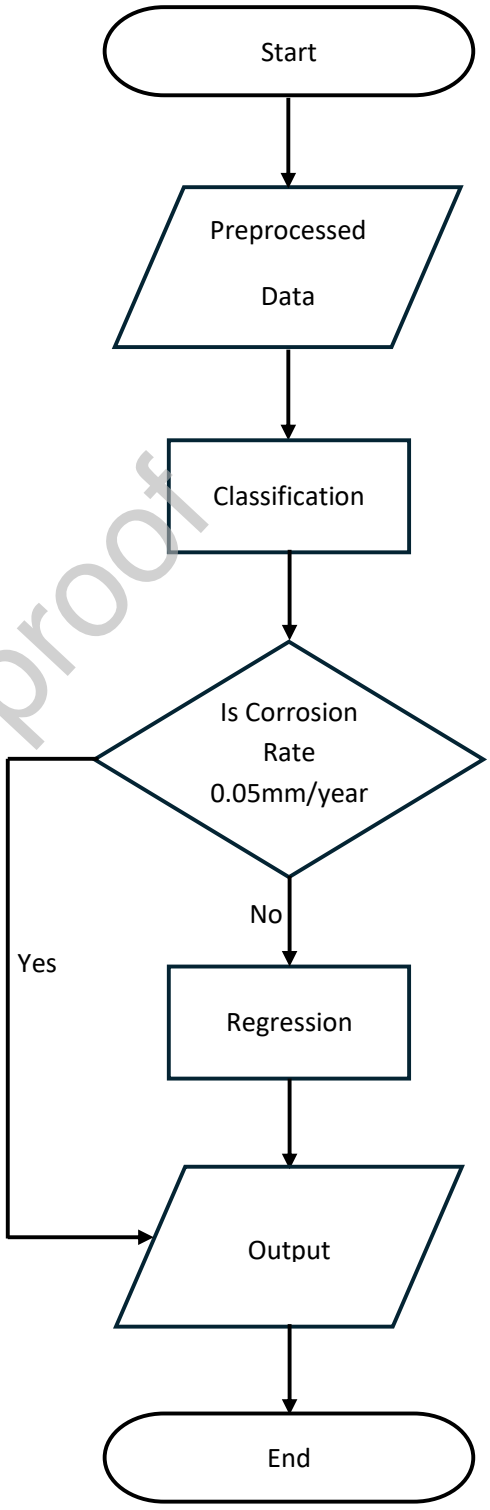
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than 0.05 mm/year. The model allows us to select a classification technique from among Random Forest classifier (RFC), k-nearest neighbors (KNN), Stochastic gradient descent classifier (SGDC), and Voting classifier (RFC+KNN+SGDC) algorithms. Regression analysis is then conducted on data that have a target value greater than 0.05 mm/year. In the regression step, the model again allows us to select a regression technique from SVM, SGD, MLP, and Voting Regressor (SVM+MLP+SGD) algorithms. Each algorithm was tested to find the best performing algorithm.

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(a) Model 1: Direct regression model



(b) Model 2: Composite classification- regression model

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Figure 5. Flowchart of the proposed algorithm for corrosion rate prediction

2.4. Tuning Hyperparameters

The grid search technique was used to tune the hyperparameters of each algorithm in both the models to enhance its performance. This entails doing an exhaustive search across a predetermined range of values to determine the ideal values for the hyperparameters of a given algorithm, such as the learning rate, regularization strength, or number of hidden layers [25]. It can take a lot of effort to refine the models using the grid search method, but doing so is vital to get the greatest results. With each approach, the hyperparameters can be carefully chosen, allowing the models to be tuned for maximum accuracy.

3. Theory

This section explains the theory behind the regression algorithms and classification algorithms employed in the present study. The mathematical background of the four regression techniques and four classification techniques are highlighted in the subsections below.

For each of the regression algorithms, we have provided the parameter learning approach to give a glimpse of the foundation of the predictive methods employed in the current study.

3.1. Support Vector Machine (SVM) Regression

The goal of a SVM regressor is to find a function $f(x)$ (equation 1) that deviates less than ε from the actual target values (y_i) for all the training datasets [26]. The SVM regressor also tries to keep the function $f(x)$ flat to achieve generalization. The SVM regression defines the linear function as:

$$f(x) = \langle w, x \rangle + b \#(1)$$

Where, w is the weight gradient, $\langle w, x \rangle$ is the dot product of the weights and input features and b is the bias intercept.

3.2. Stochastic Gradient Descent (SGD) Regression

An SGD regressor iteratively updates the model parameters to provide an objective function $E(w, b)$ with a minimum regularized training error as shown in equation 2. SGD regressor is an efficient and strong approach for large datasets [27].

$$E(w, b) = \frac{1}{n}\sum_{i=1}^{n} L\big(y_i, f(x_i)\big) + \alpha R(w) \#(2)$$

Where, $f(x_i) = w^T x_i + b$ is the linear prediction, $L\big(y_i, f(x_i)\big)$ is the loss function, $R(w)$ is the penalty term and α is the hyperparameter controlling the regularization strength.

3.3. Multi-Layer Perceptron (MLP) Regression

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An MLP regressor is an artificial neural network and a sequence of multiple transformations, suitable for complex prediction [28]. It consists of an input layer, one or more hidden layers and an output layer [29]. For every single hidden layer, the prediction $f(x)$ is given by equation (3).

$$f(x) = \sigma_{out}(W_2 \cdot \sigma_h(W_1 \cdot x + b_1) + b_2) \#(3)$$

Where, x is the input feature vector, W_1 and b_1 are the weights and biases of the hidden layer respectively, σ_h is the activation function of the hidden layer, W_2 and b_2 are the weights and biases of the output layer respectively and σ_{out} is the activation function for the output layer.

3.4. Voting Regression

The voting regressor is an ensemble technique which combines the predictions of the base models [30]. This technique does not calculate the weights separately from the independent variables, instead averages the predictions of the base models (equation 4). In the present study, it aggregates the predictions of SVM, SGD and MLP regression. The governing equation of a voting regression is given below:

$$\hat{y}_{final} = \frac{\sum_{j=1}^{m} w_j \cdot f_j(x)}{\sum_{j=1}^{m} w_j} \#(4)$$

Where, m is the number of regressors, $f_j(x)$ is the prediction for input x from the j-th model and w_j is the weight assigned to the j-th model.

As mentioned in section 2.3, model 2 classifies the dataset based on local corrosion rate prior to regression. To achieve higher accuracy, classification can be carried out using tree-based RFC, geometric distance based KNN, iteration based SGDC and voting classifier. The governing equations of each of the classifiers are detailed as follows.

3.5. Random Forest Classifier (RFC)

The RFC uses decision tree average methods to classify the dataset [31]. RFC is best suited for handling non-linear features and tabular data [32]. The RFC selects the variable and threshold that minimizes the Gini impurity (G) as shown in equation 5.

$$G = 1 - \sum_{i=1}^{c} {p_i}^2 \#(5)$$

Where, p_i is the fraction of data belonging to class i in that node. For a forest with N trees, the final classification y_{final} is the mode of all individual tree predictions $T_N(x)$ (equation 6).

$$y_{final} = mode\{T_1(x), T_2(x), \dots, T_N(x)\} \#(6)$$

3.6. k-Nearest Neighbors (KNN)

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KNN does not classify based on a training equation, instead it is an opportunity-based learner that works best for clear clusters [33]. It identifies the local neighborhood density using the distance metric $d(x, x_i)$ (equation 7).

$$d(x, x_i) = \sqrt{\sum \left(x_j - x_{i,j}\right)^2} \#(7)$$

The final classification y_{final} is done by a majority vote among the local neighbors $N_k(x)$ of an input point (equation 8).

$$y_{final} = argmax_c \sum_{i \in N_k(x)} I(y_i = c) \#(8)$$

Where, $N_k(x)$ is the set of k nearest points and I is the indicator function.

3.7. Stochastic Gradient Descent Classifier (SGDC)

SGDC works well for extremely large datasets by gradual iteration of the weights [34]. This makes it extremely fast and best suited for online learning where data appears continually. The governing equation for SGDC is same as the governing equation of SGD regressor (equation 2). By managing the loss function, SGDC provides higher variability in handling the boundary conditions [35].

3.8 Voting Classifier

The voting classifier is an ensemble technique which operates with the same approach as voting regressor (equation 4). It combines the results of RFC, KNN and SGDC by averaging their predictions and provides a robust classification. It reduces the error by achieving stability that individual classifiers are unable to achieve alone and this helps in better classification of unseen test data [36].

4. Results

The preprocessed dataset was run through two models, model 1: direct regression approach and model 2: composite classification regression approach. The results of predicted local corrosion rate obtained through regression is discussed in the subsequent sub-sections through statistical metrics such as root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE) and coefficient of determination (R^2). RMSE is the square root of the sum of square of difference between predicted value and actual value, and it penalizes the outliers in error [37]. MAE is the average of the absolute difference between the predicted value and actual value, MAPE expresses absolute error as a percentage of actual value and R^2 is the proportion of variance in the target variable that is predictable from the input features [38]. In model 2, the classification of dataset is discussed with the help of confusion matrix plotted for each technique. The confusion matrix is discussed with the assistance of statistical metrics such as accuracy, precision, recall and F1-score. Accuracy represents the ratio of correct classification predictions to the total cases examined, precision is the ratio of true positives to total predicted positives (true positive + false positive), recall is the ratio of true positive to the total true instances (true positive + false negative) and F1-score is the harmonic mean of precision and recall [39].

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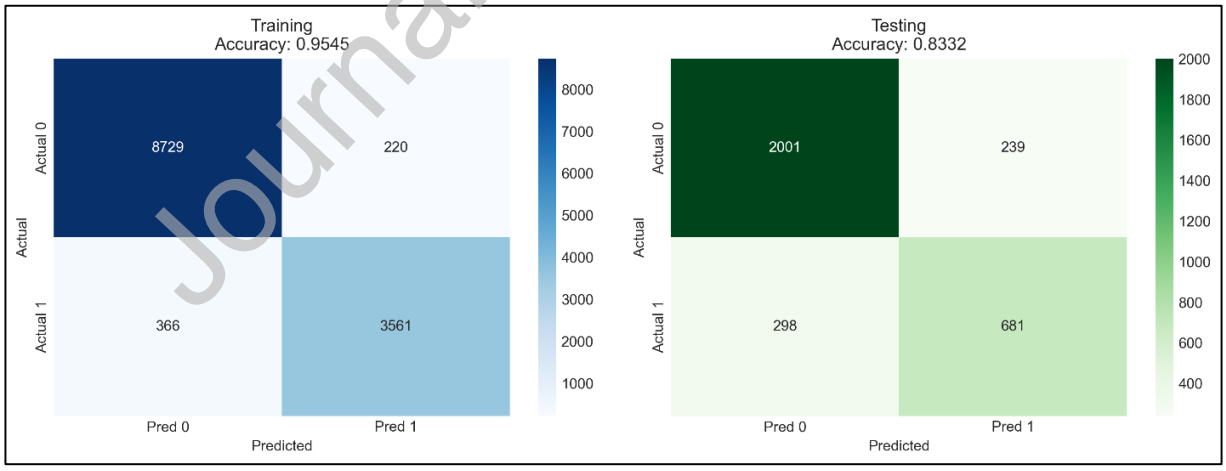
4.1. Model 1: Direct Regression Approach

The quantitative analysis between the predicted local corrosion rate and actual local corrosion rate using model 1 is collected in Table 4. The MLP regression algorithm proved to be predicting better compared with other algorithms. The RMSE and MAE is least for both training and testing using MLP regressor. Using MLP algorithm in model 1; the RMSE is 0.05162mm/year in training and 0.05209mm/year in testing, MAPE is 37.71% in training and 40.46% in testing, while R^2 is 0.346 in training and 0.226 in testing. The SVM regressor worked very poorly in prediction. It has the highest RMSE, MAE and MAPE. The negative R^2 value obtained using SVM regressor shows that SVM performed an awful job in fitting the data.

Table 4. Statistical metrics of local corrosion rate predicted by regression algorithms in model 1

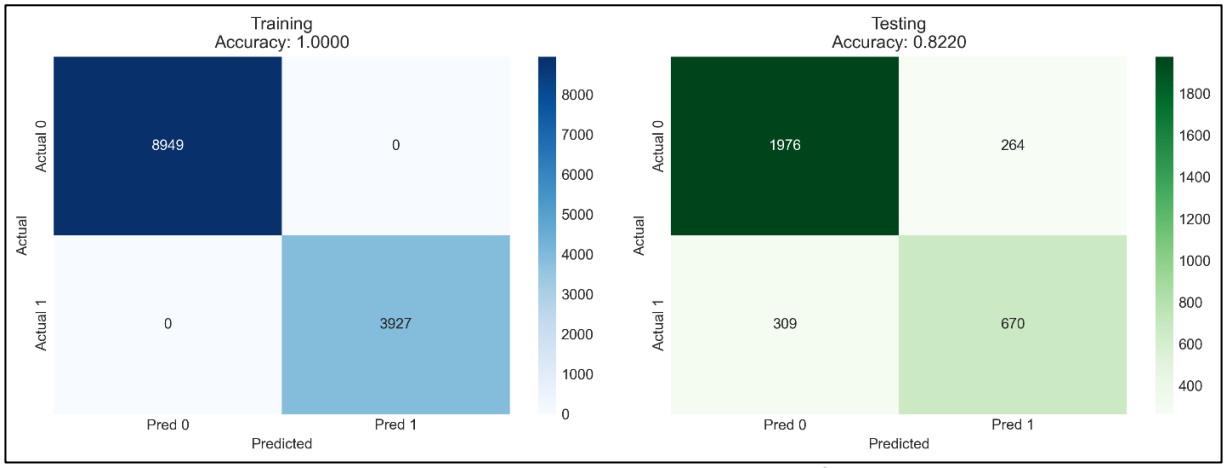
	SVM		**SGD**		**MLP**		**Voting**	
Metric	**Training**	**Testing**	**Training**	**Testing**	**Training**	**Testing**	**Training**	**Testing**
RMSE	0.07003	0.0710	0.06171	0.05766	0.05162	0.05209	0.05566	0.05432
MAE	0.05730	0.0577	0.03207	0.0311	0.02803	0.02953	0.03320	0.03358
MAPE (%)	96.876	97.640	40.1211	40.241	37.7068	40.4626	46.4990	47.9843
R^2	-0.2028	-0.4378	0.06578	0.0521	0.34632	0.22624	0.23999	0.15846

4.2. Model 2: Classification Regression Approach

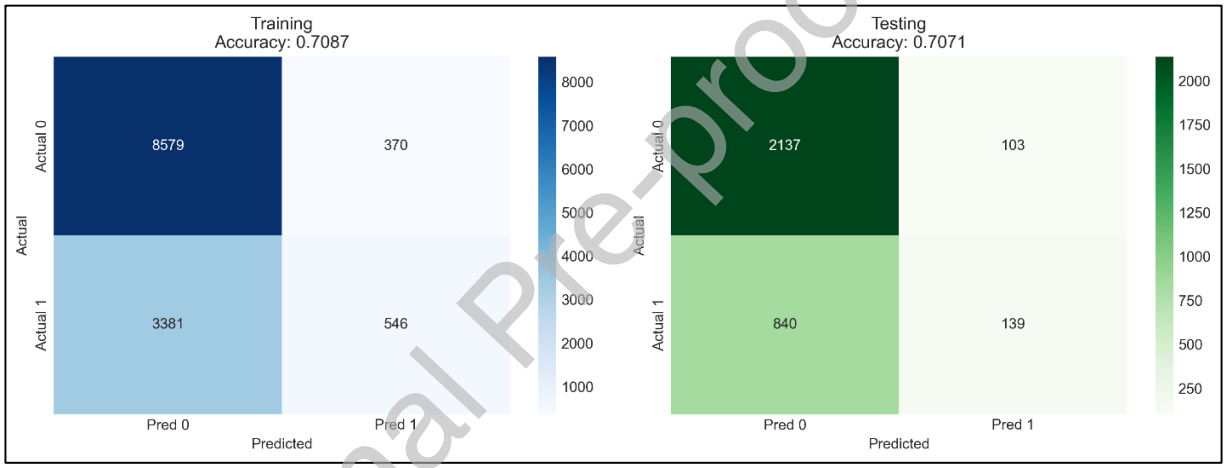


(a) Confusion matrix using RFC

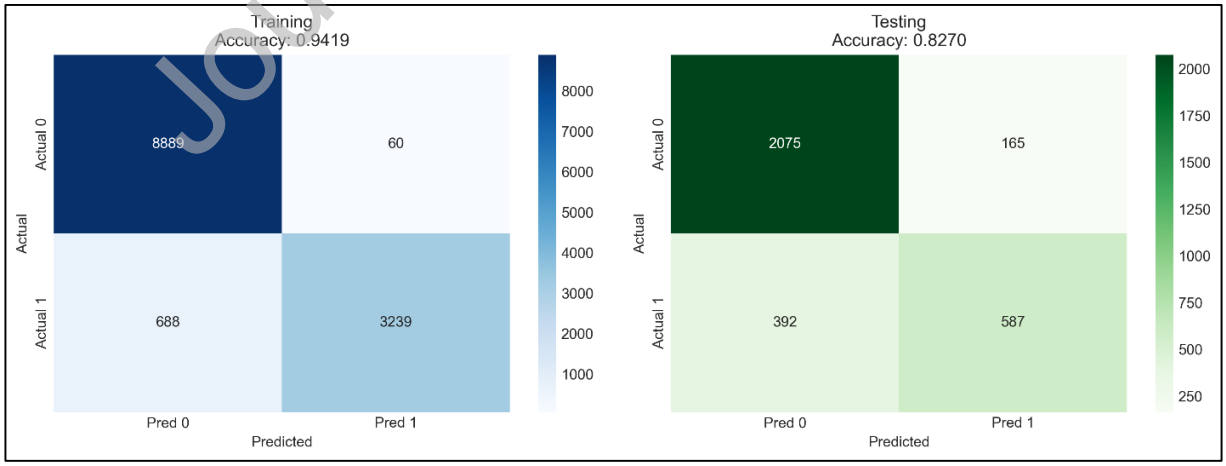
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(b) Confusion matrix using KNN



(c) Confusion matrix using SGDC



(d) Confusion matrix using voting classifier

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Figure 6. Classification of training and testing dataset using classifier algorithms. 0 is the negative class and 1 is the positive class.

In model 2, the dataset was first segregated in two classes 0 and 1 based on the local corrosion rate using classifier algorithms. The classes predicted by each of the classification algorithms along with the accuracy is shown using the confusion matrix (Figure 6). Table 5 shows the precision, recall and F1-score values of classification using each of the four classifiers. KNN classifier produced an accuracy of 1 in training but only achieved an accuracy of 0.822 in testing. The precision, recall and F1-score of KNN were all 1 in training, while in testing, voting classifier was the most precise. RFC had the highest recall of 0.6956 and highest F1-score of 0.7172 in testing.

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Table 5. Statistical metrics of confusion matrix using classification algorithms in model 2

	RFC		KNN		SGDC		Voting	
Metric	**Training**	**Testing**	**Training**	**Testing**	**Training**	**Testing**	**Training**	**Testing**
Precision	0.9418	0.7402	1	0.7173	0.5961	0.5744	0.9818	0.7806
Recall	0.9068	0.6956	1	0.6844	0.139	0.142	0.8248	0.5996
F1-Score	0.924	0.7172	1	0.7005	0.2255	0.2277	0.8965	0.6782

The quantitative analysis between the predicted local corrosion rate and actual local corrosion rate post classification using model 2 is tabulated in Table 6. The SVM regressor had the least RMSE both in training and testing, indicating that the least outlier errors were achieved using SVM regression. The MAE were lowest and very close using SVM and voting regressor. The MAPE was lowest in training when prediction was performed using voting regressor, however the MLP fared better in testing with the lowest MAPE. Yet SVM has the best R^2 both in training and testing. The MLP had the highest RMSE and MAE in both training and testing. The negative R^2 value obtained using MLP regressor shows that MLP falls short of explaining the variation in the dependent variable.

Table 6. Statistical metrics of local corrosion rate predicted by regression algorithms in model 2 post classification

	SVM	SGD	MLP	Voting

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Metric	Training	Testing	Training	Testing	Training	Testing	Training	Testing
RMSE	0.0544	0.0524	0.0571	0.0552	0.0664	0.0612	0.0559	0.0539
MAE	0.0152	0.0194	0.0163	0.0211	0.0226	0.0227	0.0157	0.0195
MAPE (%)	10.6	16.98	10.2	17.27	14.56	14.77	9.85	15.08
R^2	0.2861	0.2028	0.214	0.1184	-0.0649	-0.0847	0.2471	0.1592

5. Discussion

The statistical characteristics of the numerical dataset indicated high diversity in the spread of the data, such as operating temperature, pressure, aging, minimum measured thickness, inside diameter, water content, and pH. The distribution of local corrosion rate values showed that a significant amount of data had a corrosion rate of 0.05 mm/year. The coefficient correlation plot showed that other than fluid type, material grade and aging, other features had low linear correlation with the target. The MRMR plot showed that operating temperature had the greatest effect on the corrosion rate, which was consistent with experimental observations, but some MRMR results were inconsistent with the data scatter or measurement errors.

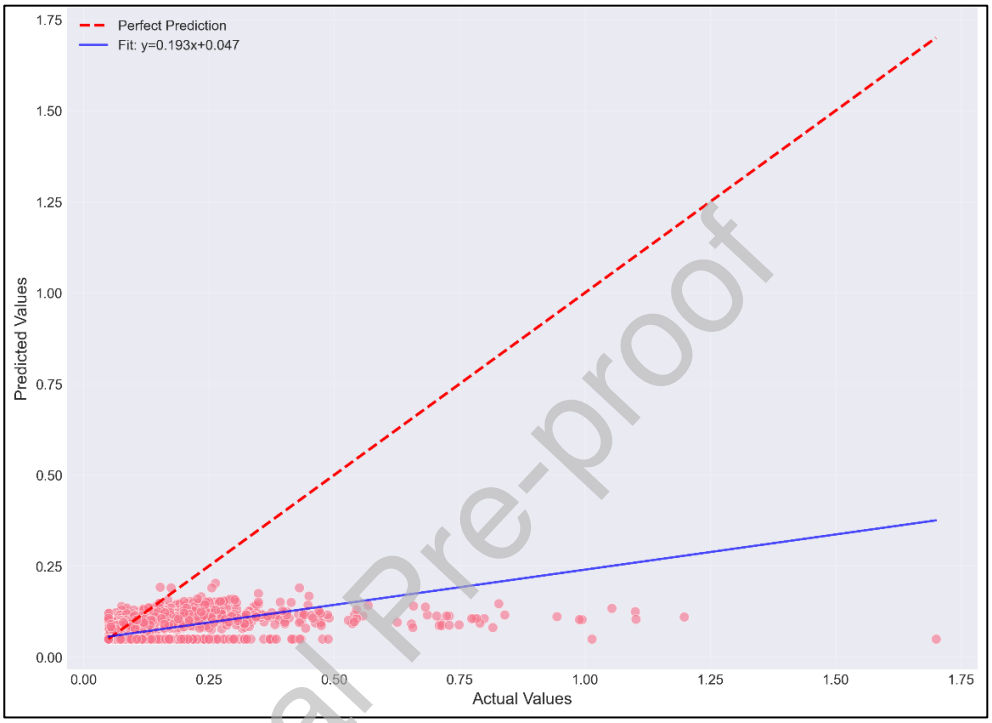
The dataset contained many duplicate data with the same or different targets, which could lead to confusion and errors in the model. A power transform was applied to make the data more Gaussian-like, which improved the model's performance by reducing heteroscedasticity and non-normality. Power transformation is a technique used in machine learning to modify the distribution of numerical features. It applies a mathematical function to the values of the feature to make it more Gaussian-like. This helps improve the performance of some machine learning models, particularly those that assume normality in the data. Power transformation can also help with dealing with duplication by making the data less susceptible to duplicate values. This is because power transformation can spread out values that are clustered together in the original distribution, reducing the likelihood of duplicates. The presence of outliers in the data caused inaccuracy in the model's predictions. However, removing them could result in data loss and bias. Therefore, removing the data should be done carefully.

Supervised learning always predicts better compared to unsupervised learning algorithms for large and diverse dataset [40]. The same was observed in the present study when model 1 based on direct regression approach (unsupervised learning) predicted the target variable with higher errors as seen from the statistical metrics. Model 2 achieved high degree of accuracy comparatively, specifically for lower corrosion rate (0.05mm/year).

5.1. Distribution and Residuals of Predicted Values Against Actual Values in Training and Testing

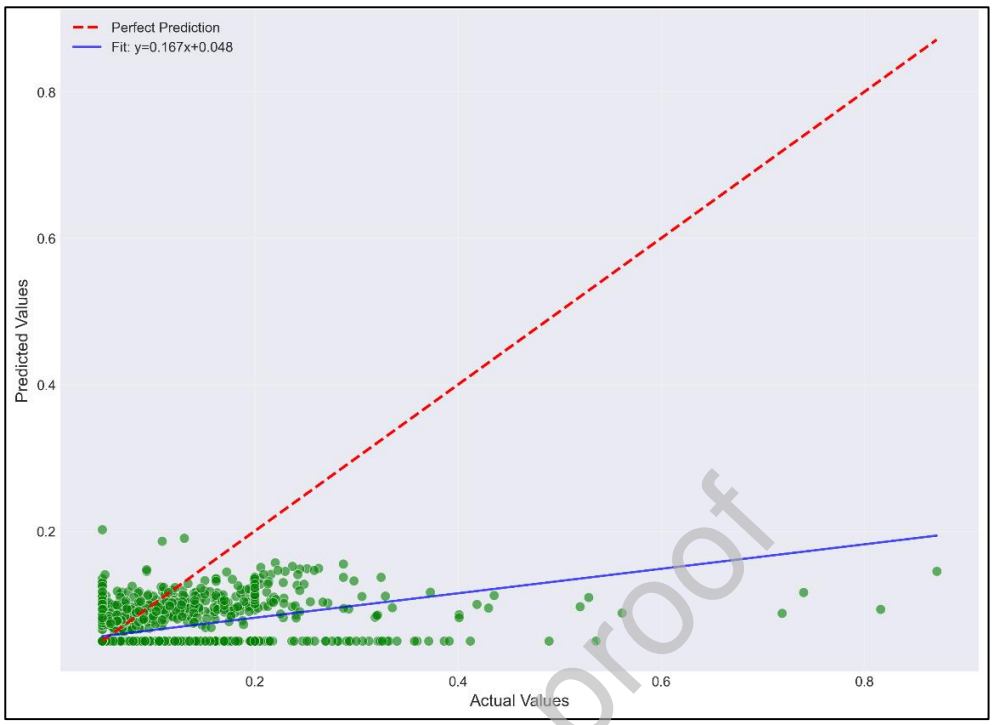
Since we achieved diversity in performance metrics using model 2, we have discussed the performance of the developed model using voting regression post classification by voting classifier in Figure 7, which represents the scatter plot of predicted values of local corrosion rate against the actual values of the

same for training and testing dataset. The dotted red line indicates the perfect fit condition (y=x) while the solid blue line is the predicted linear fit. In training, the regression fit achieved was y=0.193x+0.047, while in testing, the regression fit achieved was y=0.0167x+0.048. Even though the proximity between the predicted values and actual values of target variable at lower corrosion rate (0.05mm/year) is very close, yet the differences are large for higher corrosion rates. This indicates that the model needs some improvement for achieving better accuracy at higher corrosion rates.



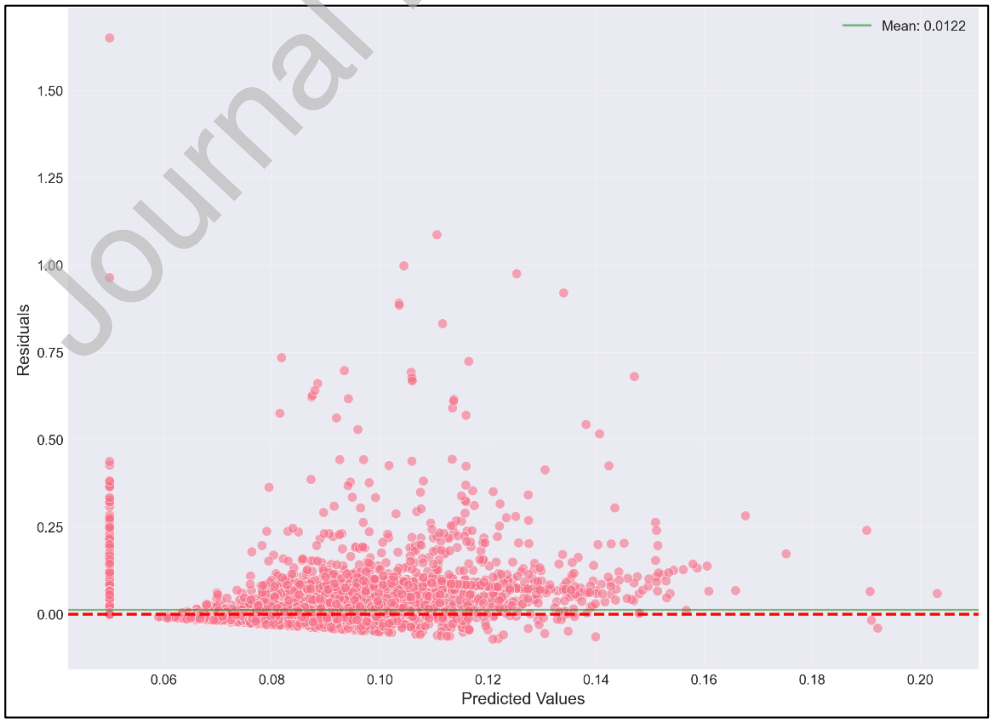
(a) Model fit for training data

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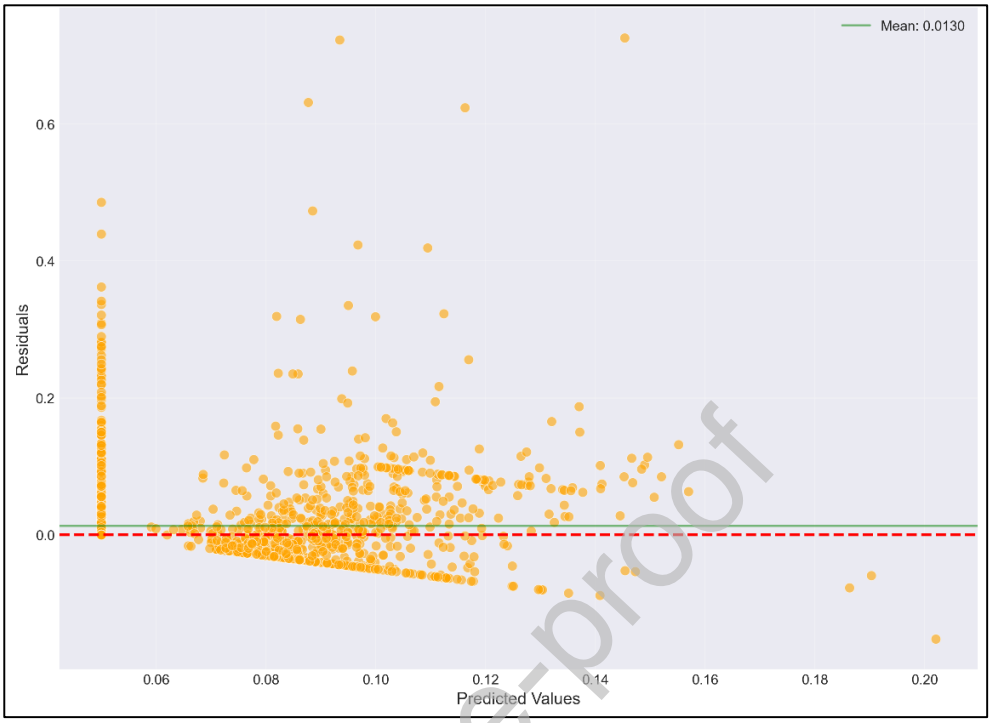
(b) Model fit for testing data

Figure 7. Correlation between predicted and actual local corrosion rate. The local corrosion rate was predicted by voting regressor after classification using voting classifier



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(a) Residuals against predicted values for the training dataset



(b) Residuals against predicted values for the training dataset

Figure 8. Scatter plot of residuals v/s predicted values for local corrosion rate. The local corrosion rate was predicted by voting regressor after classification using voting classifier

Residuals are the difference between actual values and predicted values. Figure 8 presents the scatter plot of residuals against the predicted values of the local corrosion rate in both training and testing datasets. These plots representing model errors were predicting local corrosion rate using voting regression post classification by voting classifier. The dotted red line indicates zero residual while the solid green line indicates the mean residuals in the training and testing dataset. Using voting regression after classification by voting classifier, the mean residuals in the training data was 0.0122mm/year while it was 0.0130mm/year in the testing dataset. Even though the residuals at lower corrosion rate (0.05mm/year) is rarely a non-zero value, yet the residuals are large for higher corrosion rates. This also indicates that the model falls short of achieving better accuracy for higher corrosion rates.

5.2. Uncertainty Analysis

Uncertainty is often misunderstood with risk; uncertainty is related to lack of knowledge about the consequences whereas risks are related to known probabilities [41]. Yazdi et al. [41] classified uncertainty into two types, aleatory uncertainty which is objective and occurs due to obvious nature of systems whereas epistemic uncertainty is subjective and arises due to incomplete knowledge. Yazdi [42] highlighted that effective handling of uncertainty increases the effectiveness of failure mode and effect analysis. It is evident from Yazdi's literature that uncertainty must be explicitly quantified instead of

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ignoring it. As the purpose of the present work is to reduce the accidents materializing due to failure, we have discussed the propagation of uncertainty in the prediction model as corrosion rate increases.

Bootstrap resampling technique was used in the present study to estimate the uncertainty in the model based on different corrosion rate bins (≤0.05mm/yr, 0.05-0.06mm/yr, 0.06-0.1mm/yr, 0.1-0.2mm/yr and >0.2mm/yr). Bootstrap resampling replaces a random datapoint in the new sample pool of training datasets from original datasets, while the out of bag datapoints are used for testing [43]. In the present study, 100 iterations were performed in bootstrap resampling to calculate the uncertainty in model 2. As similar patterns of uncertainty was observed for all the classifier and regressor combinations, we have discussed the uncertainty propagating in testing dataset based on different corrosion rate bins when voting classifier and voting regressor were utilized for corrosion rate prediction.

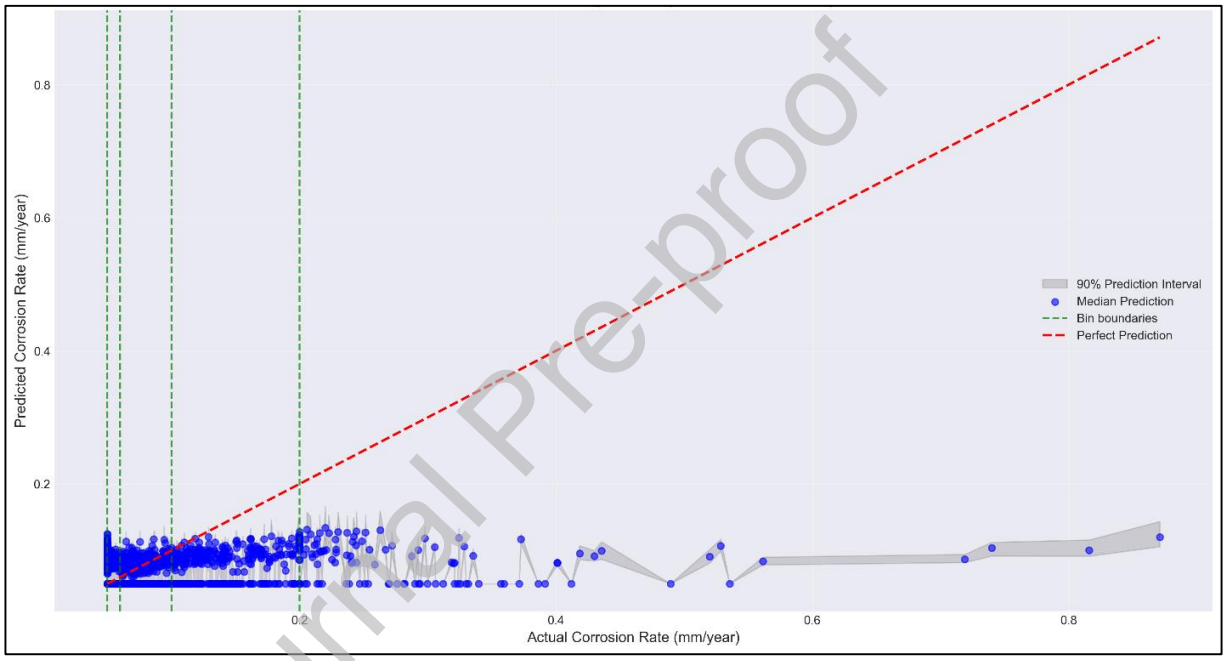


Figure 9. Prediction intervals across corrosion rate bins

Figure 9 shows the 90% predicted corrosion rate values against the actual corrosion rate values. The corrosion rate bins are classified as ≤0.05mm/yr, 0.05-0.06mm/yr, 0.06-0.1mm/yr, 0.1-0.2mm/yr and >0.2mm/yr. The median prediction as well as the 90% prediction intervals are marked accordingly. For each of the predicted values achieved after the iterations of bootstrap resampling, the median values are the 50th percentile whereas the 90% prediction intervals are the range of values from 5th percentile to 95th percentile. For corrosion rate bins ≤0.05mm/yr and 0.05-0.06mm/yr, the median of the predicted values are in proximity to the actual values. As the corrosion rate increases (0.06-0.1mm/yr, 0.1-0.2mm/yr and >0.2mm/yr), the difference between the actual values and the median values increases. The prediction interval also slides away from the perfect prediction line. Thus, the epistemic uncertainty transpiring in the model due to either limitations in the prediction models, or the imprecision in input variable, or overlooked factors, or all of them together affects the accuracy of the model.

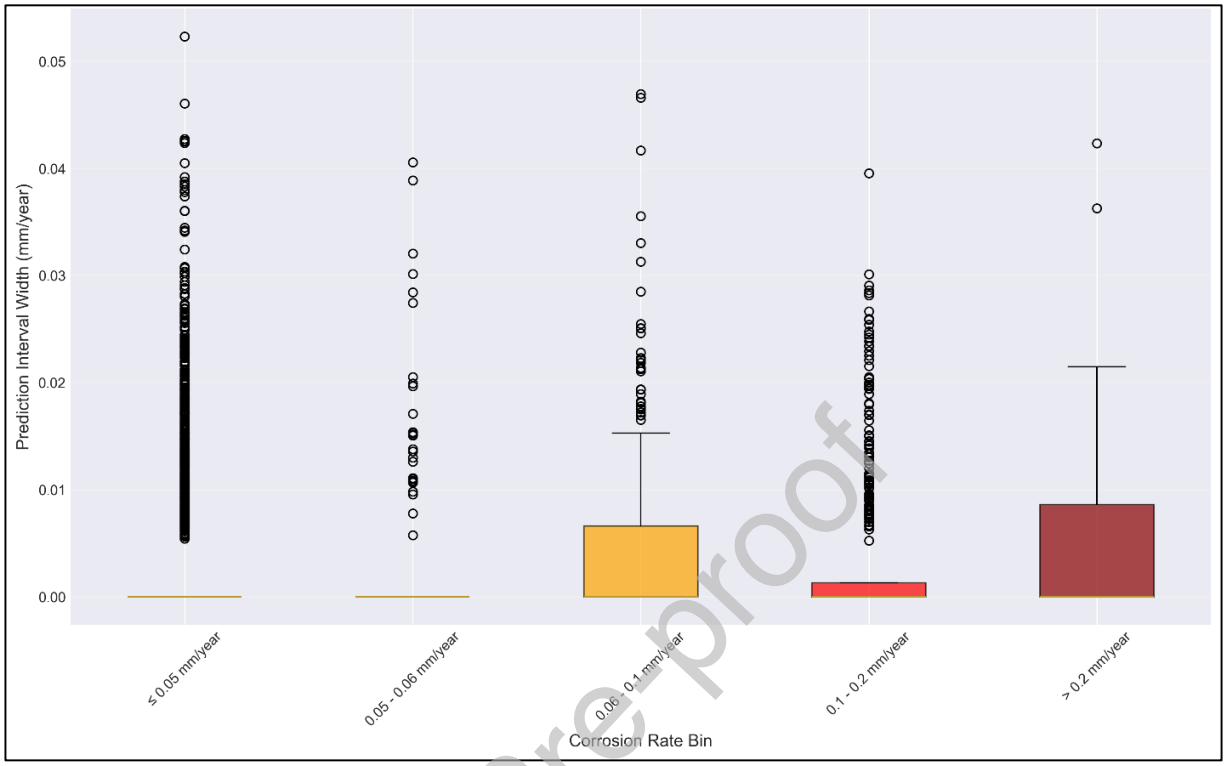


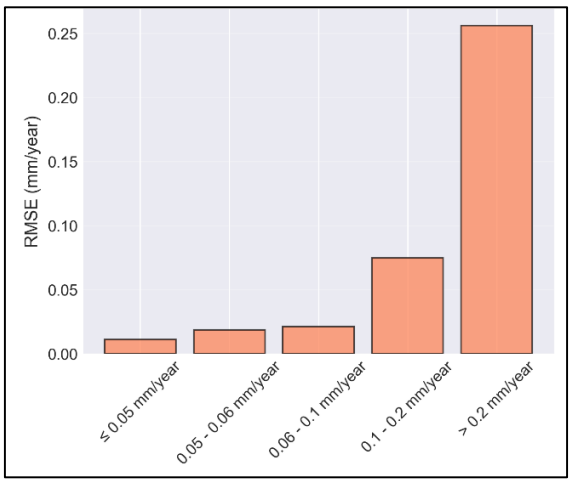
Figure 10. Uncertainty distribution by corrosion bins

The box and whisker plot (Figure 10) depicts the prediction interval width across the various corrosion rate classes (≤0.05mm/yr, 0.05-0.06mm/yr, 0.06-0.1mm/yr, 0.1-0.2mm/yr and >0.2mm/yr). For corrosion rates ≤0.05mm/yr, the uncertainty is 0 whereas the interquartile range is at most 0.04mm/yr. Many outliers are also observed within the corrosion rates of ≤0.05mm/yr as the interquartile range is quite small. In the corrosion rate bin of 0.05-0.06mm/yr, the uncertainty rises to 0.02mm/yr with interquartile range being the same as that for ≤0.05mm/yr. As the corrosion rate increases to 0.06-0.1mm/yr, the box widens with median uncertainty 0.12mm/yr and interquartile range extends to 0.15mm/yr. The same pattern is observed for higher corrosion rates of 0.1-0.2mm/yr and >0.2mm/yr with uncertainty reaching 0.22mm/yr and 0.31mm/yr respectively.

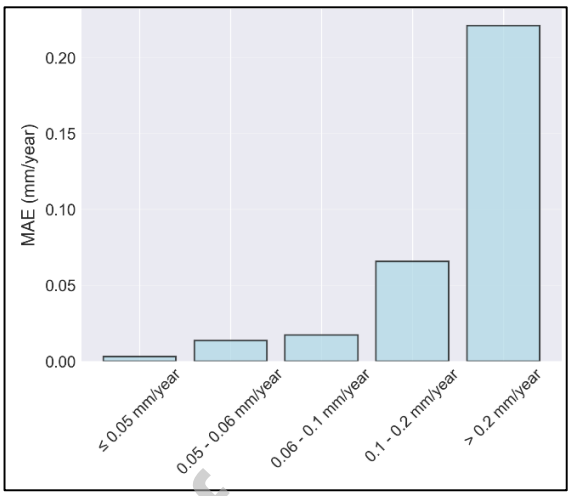
5.3. Targeted Performance Metrics

By segregating the testing set according to the corrosion rate bins (≤0.05mm/yr, 0.05-0.06mm/yr, 0.06-0.1mm/yr, 0.1-0.2mm/yr and >0.2mm/yr), we evaluated the performance metrics RMSE, MAE, MAPE, R^2 and prediction bias for each corrosion rate bins. The metrics discussed are calculated for predicted corrosion rate using voting classifier and voting regressor in model 2. Prediction bias is the systematic difference between the average predicted values and average truth value.

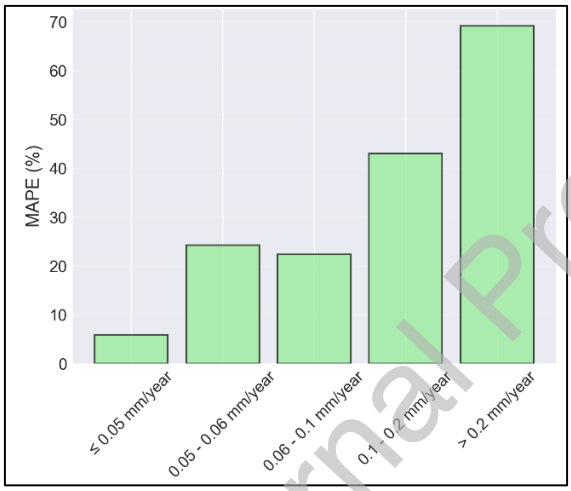
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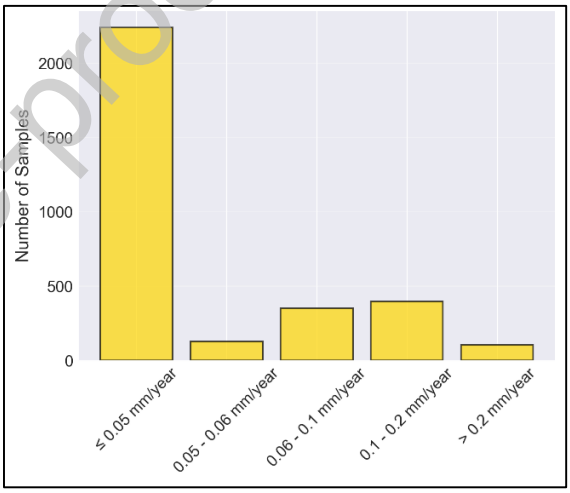
(a) RMSE



(b) MAE

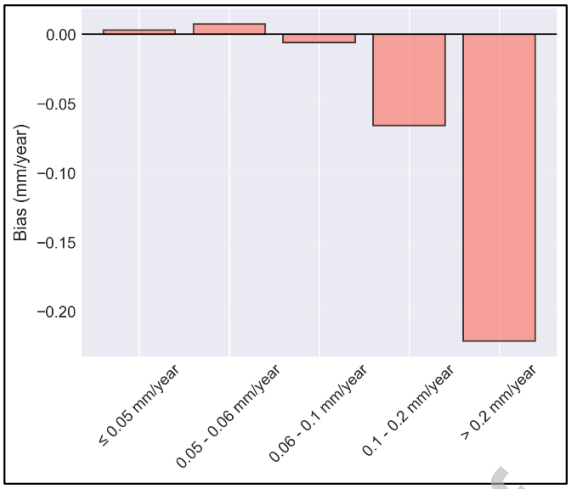


(c) MAPE



(d) Sample distribution

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(e) Prediction bias

Figure 11. Targeted performance metrics evaluated as per corrosion rate bins

The targeted performance metrics against different corrosion rate bins are plotted (Figure 11). In testing dataset, out of 3219 datapoints, the actual corrosion rate values of 2240 datapoints are ≤0.05mm/yr. Only the actual corrosion rate values of 105 datapoints are >0.2mm/yr. For corrosion rates between 0.05-0.06mm/yr, 0.06-0.1mm/yr, 0.1-0.2mm/yr, there lies 127, 351 and 396 datapoints respectively. The RMSE, MAE and MAPE in corrosion rate bin ≤0.05mm/yr are 0.011mm/yr,0.0029mm/yr and 5.9% whereas the same values are 0.26mm/yr, 0.22mm/yr and 68.2% in corrosion rate bin >0.2mm/yr respectively. The errors in corrosion rate bins of 0.05-0.06mm/yr and 0.06-0.1mm/yr increases gradually and it increases sharply for corrosion rates >0.1mm/yr. The prediction bias for corrosion rates ≤0.05mm/yr is 0.0029mm/yr while the same is -0.22mm/yr for corrosion rates >0.2mm/yr. The R^2 evaluates to 0.427 for corrosion rate ≤0.05mm/yr while it reaches -1 for corrosion rates >0.2mm/yr which shows that the model is unsuitable for corrosion rates >0.2mm/yr. Based on the targeted performance metrics, we have suggested the correction factors in subsection 5.5 that can be incorporated together with this model for the time being the model is improved and added with new datapoints at higher corrosion rate.

5.4. Industrial Challenges

With respect to the availability of open-source real-time datasets, the competition among the industry players in the oil and gas industry is very stiff. Memorandums, regulations and agreements prevent the data from being shared as open source. This is one of the reasons that research articles published in the past lack the quantity of real-time datasets. For the same reason, the use of simulated datasets is found to be in abundance in this domain. However, looking at the complex nature of surroundings and environment in which piping systems are operating, the real-time data will always hold precedence over simulated data.

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Multiple internal corrosion mechanisms can affect the piping systems such as Amine Corrosion, Ammonium Bisulfide Corrosion, Ammonium Chloride and Amine Hydrochloride Corrosion, Aqueous Organic Acid Corrosion, Brine Corrosion, Caustic Corrosion, CO_2 Corrosion, Concentration Cell Corrosion, Erosion/Erosion-Corrosion, Flue Gas Dew Point Corrosion, High-temperature H_2/H_2S Corrosion, Hydrochloric Acid Corrosion, Hydrofluoric Acid Corrosion, Microbiologically Influenced Corrosion, Naphthenic Acid Corrosion, Oxygenated Process Water Corrosion, Phenol (Carbolic Acid) Corrosion, Phosphoric Acid Corrosion, Sour Water Corrosion (Acidic) and Sulfuric Acid Corrosion [14]. Thus, having a sound knowledge of factors affecting internal corrosion in piping system becomes mandatory for developing a soundly accurate model to predict internal corrosion in the piping system. The survival and dependability of an O&G platform is affected by corrosion prediction as huge economic losses are attached with outcomes of corrosion [44]. The chemical and physical parameters involved in the O&G operations play an important role in the spread of corrosion [45,46]. Hence, it becomes crucial to understand the factors associated with internal corrosion in the field of asset integrity management. The better knowledge of factors affecting corrosion had led to better prediction of corrosion spread in the past either using experience models, semi-empirical models, mechanism models or data-driven models.

5.5. Future Recommendations

For future studies, the present model can be refined using more accurate ensemble learning algorithms such as extreme gradient boosting (XGBoost), adaptive boosting (AdaBoost) and bootstrap aggregating (bagging) along with Bayesian optimization (BO) algorithms [10, 11]. The model can also improve when more diverse dataset at higher corrosion rate is added to the present dataset and the uncertainty in the model decreases. However, for now, based on the uncertainty calculation, it is recommended to consider a correction factor to compensate for potential underprediction of corrosion rates. Table 7 shows a preliminary suggestion on correction factors to be used for corrosion rates. This suggestion is based on testing various cases with different algorithms and uncertainty propagation in the model across the different corrosion bins. It is highly recommended to periodically review this table with input from a subject matter expert to refine the correction factors.

Table 7. Suggested correction factors for predicted values of local corrosion rate (CR)

Estimated ML CR (mm/year)	**Note**	**Observation/ Recommendation**
CR=0.05	Most accurate	Apply model's typical error margin as a safety factor of **1.2**
CR<0.06	Slight overprediction may occur	Apply model's typical error margin as a safety factor of **1.2**
CR (0.06 and 0.1)	Slight underprediction may occur	Apply a safety factor of **1.5**
CR (0.1 and 0.2)	Underprediction may occur	Model not accurate, apply a safety factor of **2**
CR (>0.2)	Extreme Underprediction may occur	Model not accurate, seek expert advice to choose a conservative safety factor

6. Conclusion

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In summary, the analysis of the corrosion dataset revealed several important insights for building a machine learning model. The data showed significant variability and scatter, with some features having low correlation with the target and some data points having inconsistent or duplicate values. Despite the challenges seen in the datasets, a reasonable machine learning model developed to predict annual corrosion rate for a given 12 inputs (features). To enhance the model's performance, it is necessary to reduce the impact of duplicate and inconsistent data. By applying appropriate data preprocessing techniques and feature selection methods, the model was improved and is now able to capture the underlying patterns and predict the local corrosion rate with higher accuracy. Generally, machine learning models are not perfect and inherently have some degree of error. This error can be due to a variety of factors, such as the quality and quantity of the training data, the complexity of the model, or the algorithm used for learning. The study has produced a machine learning model that achieves a good accuracy for lower corrosion rate (0.05mm/year). However, it is essential to acknowledge that this model may not suffice for higher corrosion rates, and there are certain limitations to the model's performance. To improve model accuracy, we had also discussed the use of correction factors, model refinement and reducing bias nature of dataset by adding more diverse values at higher corrosion rate.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The datasets analyzed during the current study are not publicly available due to legal, contractual and confidentiality restrictions imposed by third-party licensing agreements and privacy laws.

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Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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